# ReticleOverlay

## User Guide

v2026.05.12.002

12 reticle shapes • Dual-layer rendering • Multi-monitor • Hide from capture

Windows 10/11 • .NET 10 • WPF/MVVM

# 1. Overview

ReticleOverlay renders a fully customisable crosshair or reticle on top of all other windows using a layered, click-through transparent overlay. It supports 12 built-in shapes, a two-layer independent rendering system, per-monitor positioning, capture exclusion, and 5 hotkey-accessible presets.

Designed for: competitive gaming, video production, accessibility (visual anchor), accessibility screen recordings, and streaming where a persistent centre marker improves usability.

# 2. System Requirements

| Component | Requirement |
|---|---|
| OS | Windows 10 v1903 or Windows 11 |
| Runtime | .NET 10 Desktop Runtime (x64) |
| RAM | 32 MB minimum |
| Display | Any resolution; DPI-aware per-monitor |
| DirectX | DX9 or later (for overlay compositing) |

# 3. Installation

Run the installer or use the portable build. No elevation required. Settings stored in %APPDATA%\ReticleOverlay\settings.json.

# 4. Interface Tour

| Zone | Location | Purpose |
|---|---|---|
| 1 Identity/Scale | Top bar | App name, version, DPI scale |
| 2 NavRow | Row 2 | Shape \| Colours \| Position \| Layer2 \| Presets \| Settings |
| 3 Toggle/Preview | Row 3 | Master toggle (F8) + live reticle preview canvas |
| 4 Main / Sliders | Centre | Shape picker, size/thickness/gap sliders, rotation dial, colour palette |
| 5 Presets 1–5 / Save | Bottom bar | Preset slots; F9/F10 cycle; resize grip bottom-right |

# 5. Reticle Shapes

Select a shape in the Shape panel (NavRow → Shape). All 12 shapes respect Size, Thickness, CentreGap, and Rotation:

| Shape | Description |
|---|---|
| Crosshair | Classic ± cross — the default |

| | |
|---|---|
| Dot | Filled circle; size controls diameter |
| Circle | Hollow ring; thickness controls stroke weight |
| CrossDot | Crosshair with centre dot |
| T-Shape | Three-arm T; useful for asymmetric aim |
| Chevron | V-shape pointing up; gap = chevron opening |
| Diamond | Four-point rotated square |
| Arrow | Single directional pointer; rotation sets heading |
| Hexagon | Six-sided polygon outline |
| Octagon | Eight-sided polygon outline |
| CircleCross | Circle with inscribed crosshair |
| CustomPNG | Load a PNG file; alpha channel respected, size scaled |

■ CustomPNG: recommended size 128×128 px. Transparent pixels are fully click-through.

# 6. Shape Parameters

| Parameter | Range | Default | Description |
|---|---|---|---|
| ReticleSize | 4–200 px | 32 px | Overall bounding size of the reticle |
| ReticleThickness | 1–20 px | 2 px | Stroke/arm width |
| CentreGap | 0–49% | 10% | Dead-zone radius as % of size; arms start here |
| Rotation | 0–359° | 0° | Clockwise rotation applied to entire shape |
| OffsetX | -400 to +400 px | 0 | Horizontal shift from monitor centre |
| OffsetY | -400 to +400 px | 0 | Vertical shift from monitor centre |

# 7. Two-Layer System

Layer 1 is the primary reticle. Layer 2 is a fully independent second reticle rendered in the same overlay window. Enable Layer 2 in NavRow → Layer2.

Each layer has its own: Shape, Size, Thickness, CentreGap, Rotation, Colour, OffsetX/Y, Opacity, and Visibility toggle.

Common use cases:

• Layer 1: large circle for situational awareness + Layer 2: small dot for precision aim.

• Layer 1: crosshair in primary colour + Layer 2: outline in contrasting colour for visibility on varied backgrounds.

■ Both layers share the same HideFromCapture and MonitorIndex settings.

# 8. Colour & Palette

Each layer has a primary colour and an optional secondary colour for two-colour gradient rendering. Click the colour swatch to open the colour picker.

The 8-slot colour palette below the picker stores favourite colours for fast reuse. Click a slot to apply; right-click to assign the current colour to that slot.

| Colour Setting | Description |
|---|---|
| Primary Colour | Main reticle stroke/fill colour |
| Secondary Colour | Used for outline mode or gradient; set alpha=0 to disable |
| Opacity | 0–100% — independent of colour alpha; affects entire layer |

# 9. Position & Offset

By default the reticle centres on the selected monitor. OffsetX and OffsetY (–400 to +400 px) shift it from centre. Use the nudge hotkeys for fine control:

| Hotkey | Action |
|---|---|
| Ctrl+Alt+Arrow | Nudge reticle 1 px in direction |
| Ctrl+Alt+Shift+Arrow | Nudge reticle 10 px in direction |
| Ctrl+Alt+Home | Reset offset to 0,0 |

# 10. Multi-Monitor Support

MonitorIndex (0 = primary, 1 = second, etc.) pins the overlay to a specific display. The overlay tracks that monitor's position and size automatically, including when monitors are added or removed.

Per-preset monitor assignments allow switching between display configurations with a single hotkey press (F9/F10).

# 11. Hide from Capture

When HideFromCapture is enabled (Settings → Hide from Capture), ReticleOverlay calls SetWindowDisplayAffinity(WDA_EXCLUDEFROMCAPTURE). This prevents the reticle from appearing in:

- OBS / Streamlabs screen capture sources
- Windows Game Bar recordings
- Screenshot tools (PrintScreen, Snipping Tool)
- Remote desktop / RDP sessions

■ The reticle remains visible on the physical monitor but is invisible to any capture software.

# 12. Hotkeys

| Hotkey | Action | Configurable |
|---|---|---|

| | | |
|---|---|---|
| F8 | Toggle overlay on/off | ✓ |
| F9 | Load next preset (cycles 1→5→1) | ✓ |
| F10 | Load previous preset | ✓ |
| Ctrl+Alt+↑↓←→ | Nudge reticle 1 px | ✓ |
| Ctrl+Alt+Shift+↑↓←→ | Nudge reticle 10 px | ✓ |
| Ctrl+Alt+Home | Reset offset | ✓ |
| Ctrl+Shift+1–5 | Load specific preset | ✓ |

- Hotkeys are registered via RegisterHotKey (Win32) and work globally when overlay is active.
- Modifier syntax: Ctrl / Alt / Shift / Win separated by +. Example: Ctrl+Alt+F8

# 13. Presets

Five preset slots capture: Shape, all shape parameters, Layer 2 configuration, primary and secondary colours, 8-slot palette, OffsetX/Y, MonitorIndex, HideFromCapture, and all hotkey assignments.

Cycle presets with F9/F10 or load directly with Ctrl+Shift+1–5.

# 14. Undo / Redo

All parameter changes (sliders, colour picker, dropdowns) are undoable. Undo depth: 50 steps.

| Shortcut | Action |
|---|---|
| Ctrl+Z | Undo last change |
| Ctrl+Y | Redo |

- Undo history is per-session and does not persist across application restarts.

# 15. Settings Reference

| Setting | Type | Default | Description |
|---|---|---|---|
| ReticleType | Enum | Crosshair | One of 12 shape names |
| ReticleSize | Int | 32 | 4–200 px |
| ReticleThickness | Int | 2 | 1–20 px |
| CentreGap | Float | 0.10 | 0–0.49 as fraction of size |
| Rotation | Int | 0 | 0–359 degrees |
| OffsetX / OffsetY | Int | 0 | ±400 px from monitor centre |
| MonitorIndex | Int | 0 | 0 = primary monitor |

| | | | |
|---|---|---|---|
| HideFromCapture | Bool | false | Exclude from screen capture |
| UiScale | Enum | 1.0 | 1.0/1.25/1.5/2.0 |
| ThemeName | String | Dark | One of 20 themes |
| ThemeAuto | Bool | false | Follow Windows mode |

# 16. Themes

All 20 themes available: Dark, Light, Nord, Dracula, Catppuccin Mocha, Catppuccin Latte, Ayu Dark, Ayu Light, Ayu Mirage, Cobalt2, GitHub Light, Gruvbox Dark, Monokai, One Dark, Rose Pine, Solarized Dark, Solarized Light, Tokyo Night, Warm Light, High Contrast.

# 17. Accessibility

| Criterion | Level | Status |
|---|---|---|
| 1.4.6 Contrast | AAA | Control panel UI ≥7:1 on all themes |
| 2.5.5 Target Size | AA | All controls ≥44×44 px |
| 2.4.7 Focus Visible | AA | 2 px Gold/Cyan ring |
| 1.4.1 Use of Colour | A | Active preset indicated by label, not colour alone |
| 2.1.1 Keyboard | A | Full keyboard access including nudge hotkeys |

# 18. Troubleshooting

| Symptom | Fix |
|---|---|
| Reticle not visible | Check master toggle (F8); verify MonitorIndex matches active display |
| Reticle appears in OBS | Enable HideFromCapture in Settings |
| CustomPNG not loading | PNG must be RGBA; max recommended 512×512 px |
| Hotkey not registering | Another app has registered the same key — reassign in Settings → Hotkeys |
| Overlay flickers on game launch | Game running exclusive fullscreen; switch game to borderless windowed |
| Offset resets on restart | Click Save or Ctrl+Shift+S before closing to persist to preset |

# 19. Build Notes

Version v2026.05.12.002 built with .NET 10, C# 14, WPF. Overlay uses WS_EX_LAYERED + WS_EX_TRANSPARENT + WS_EX_TOPMOST. HideFromCapture via SetWindowDisplayAffinity(WDA_EXCLUDEFROMCAPTURE). Hotkeys via RegisterHotKey hooked through HwndSource. Versioning: yyyy.MM.dd.NNN.

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